

HINT AND CONQUER: NEW QUEEN DOMINATION NUMBERS BY SAT SOLVING

Abstract

The queen domination problem asks for the minimum number of queens required to dominate all squares of an $n \times n$ chessboard. For even n with $26 \leq n \leq 120$, it has been known since 2001 that this number is either $n/2$ or $n/2 + 1$, but apart from $n = 30$ the exact value has remained open. In this paper, we revisit a classical counting argument used to establish lower bounds for queen domination and reformulate it into constraints that can be efficiently encoded in SAT. Our approach partitions the search space according to the outermost empty rows and columns of a dominating set, which define a box. For each box, we generate a SAT formula augmented with counting hints derived from the argument. These hints impose strong restrictions on the possible coverage of the box and substantially improve solving performance compared with the same encoding without them. The resulting approach also outperforms the previous SAT encoding for queen domination on this task. Using this method, we establish six new domination numbers: $\gamma(Q_{26}) = 14$, $\gamma(Q_{28}) = 15$, $\gamma(Q_{32}) = 17$, $\gamma(Q_{34}) = 18$, $\gamma(Q_{36}) = 19$, and $\gamma(Q_{38}) = 20$. Together with the previously known values, this determines the domination number for every even $n \leq 38$. For every unsatisfiable instance, the SAT solver generates a DRAT proof, which is independently checked using `drat-trim`.

1 Introduction

The queen domination problem asks for the minimum number of queens needed to dominate an $n \times n$ chessboard, where a queen dominates all squares in its row, column, diagonal, and anti-diagonal. This minimum number is denoted by $\gamma(Q_n)$.

For even n , the lower bound $\gamma(Q_n) \geq (n-1)/2$, established independently by Raghavan and Venkatesan [7] and Spencer (see [2]), implies $\gamma(Q_n) \geq n/2$. On the other hand, the constructions of Östergård and Weakley [6] show that $\gamma(Q_n) \leq n/2 + 1$ for every even n with $26 \leq n \leq 120$. Thus, for these boards, the domination number is either $n/2$ or $n/2 + 1$. Apart from $\gamma(Q_{30}) = 15$, the exact values were unknown [10]. These cases have thus been open for more than two decades. Determining them therefore reduces to deciding whether $n/2$ queens can dominate the board.

Recently, Rostami and Bright [8] introduced a SAT-based approach to queen domination and enumerated all minimum dominating sets up to symmetry for boards of size $n \leq 19$. Their method was designed primarily for exhaustive enumeration, rather than specifically for proving that a given number of queens is insufficient. This motivates investigating whether a more specialized SAT encoding can solve such instances more efficiently and settle some long-standing open cases.

Our approach is inspired by the work of Subercaseaux et al. [9], who showed that adding clique hints substantially improves SAT performance in queen graph coloring. These hints encode consequences of pigeonhole-principle reasoning that SAT solvers struggle to derive on their own. A similar opportunity may arise in queen domination. We revisit the counting argument behind the lower bound $\gamma(Q_n) \geq (n-1)/2$ [7, 2], conjecturing that SAT solvers struggle to derive the consequences of this argument on their own, and investigate whether encoding them explicitly can similarly improve SAT solving. Our main challenge is to reformulate the argument into

constraints that can be encoded efficiently. To this end, we partition the search space according to the outermost empty rows and columns, which define a box for each dominating set. For each box, we generate a SAT formula augmented with counting hints derived from the argument.

For proving that a given number of queens is insufficient, our experiments show that the counting hints substantially improve solving performance compared with the same encoding without hints. For the same task, the resulting encoding also outperforms that of Rostami and Bright [8]. Using the proposed approach, we establish six new domination numbers:

$$\gamma(Q_{26}) = 14, \quad \gamma(Q_{28}) = 15, \quad \gamma(Q_{32}) = 17, \quad \gamma(Q_{34}) = 18, \quad \gamma(Q_{36}) = 19, \quad \gamma(Q_{38}) = 20.$$

Together with the previously known values, this determines $\gamma(Q_n)$ for every even $n \leq 38$. Each of these cases is settled with about one hour of solving time or less. All unsatisfiability results are independently verified using a proof checker.

Section 2 develops the counting argument. Section 3 presents its SAT encoding. Section 4 describes the implementation and validation. Sections 5 and 6 report the new domination numbers and evaluate the impact of the counting hints. Finally, Section 7 concludes the paper.

Related work. The queen domination problem has been studied both theoretically and computationally. Theoretical investigations have established lower and upper bounds on $\gamma(Q_n)$, including bounds that differ by only one queen for certain board sizes [7, 6, 10]. Related work has also investigated the independent domination number and asymptotic bounds based on special families of dominating sets.

Computational approaches have been developed to determine exact domination numbers and enumerate minimum dominating sets. Gibbons and Webb [3] incorporated isomorphism rejection into backtracking search. Kearse and Gibbons [4] investigated complementary search strategies based on queen placements, undominated squares, and attacking sets. More recently, Bird [1] developed the specialized UNIDOM solver, which combines branching heuristics, bounding techniques, and distributed computation to enumerate minimum dominating sets. Rostami and Bright [8] subsequently introduced a proof-producing SAT approach and enumerated all minimum dominating sets up to symmetry for $n \leq 19$.

SAT solving has also been applied to other problems on the queen’s graph. Lauri [5] investigated its domatic number, while Subercaseaux et al. [9] studied SAT encodings for its chromatic number. The latter demonstrated the importance of encoding choices, such as clique hints, for solving performance. Our work has a similar motivation but focuses on queen domination, using counting hints derived from classical lower-bound arguments.

2 The Counting Argument

This section revisits the counting argument used by Raghavan and Venkatesan [7] and Spencer (see [2]) to prove $\gamma(Q_n) \geq (n-1)/2$. Their argument considers the outermost empty rows and columns of a dominating set and counts how the squares on these lines can be covered. We refine it into an identity (Lemma 1), which is the form needed for the SAT encoding.

We number the columns of the $n \times n$ board $1, \dots, n$ from left to right and the rows $1, \dots, n$ from bottom to top, and we write (x, y) for the square in column x and row y , so that $(1, 1)$ is the bottom-left corner. A *diagonal* consists of all squares with the same value of $x - y$, and an *anti-diagonal* of all squares with the same value of $x + y$. A *line* is a row, column, diagonal, or anti-diagonal of the board, so every square lies on exactly four lines. Let D be a dominating set of γ queens. A line is *active* if it contains at least one queen of D , and *empty* otherwise. For $n \geq 3$, the queens on the squares (i, i) with $i \notin \{1, 3\}$ dominate the board, so $\gamma(Q_n) \leq n - 2$ and we assume $\gamma \leq n - 2$ throughout. Since D then occupies at most $n - 2$ columns and at most $n - 2$ rows, it leaves at least two columns and at least two rows empty. Let a and b be the

leftmost and rightmost empty columns, and let c and d be the lowest and highest empty rows. The *box* of D is the rectangle of squares (x, y) with $a \leq x \leq b$ and $c \leq y \leq d$, as illustrated in Figure 1. It follows directly from the definition that the four sides of the box (columns a, b and rows c, d) are empty, while every column to the left of a or to the right of b , and every row below c or above d , is active.

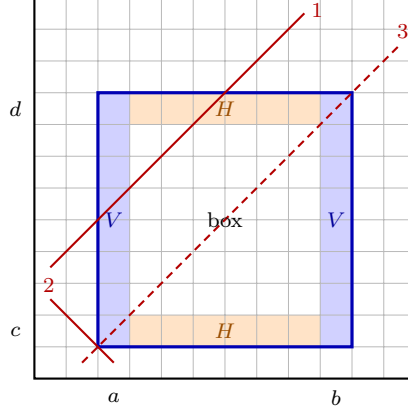


Figure 1: The box of a set of queens (queens not shown). Columns a, b and rows c, d are empty, and every row and column outside the box contains a queen. The vertical sides V (blue) and the horizontal sides H (orange) are shaded while the corners belong to both. Diagonal 1 meets V once, anti-diagonal 2 misses V , and diagonal 3, a corner-to-corner diagonal of a square box, meets V twice.

Let us call the squares of the box in columns a and b its *vertical sides* V , and the squares in rows c and d its *horizontal sides* H . Thus, V contains $2(d - c + 1)$ squares and H contains $2(b - a + 1)$ squares, with the four corners belonging to both.

Every square of V lies in an empty column, so it can only be covered by its row, its diagonal, or its anti-diagonal. In total, V has $2(d - c + 1)$ squares and each needs to be covered.

An active row covers two squares of V if it lies strictly between rows c and d , and none otherwise. The $n - 1 - (d - c)$ rows outside the box are all active, and with γ queens there are at most γ active rows in total. So at most $\gamma - (n - 1 - (d - c))$ active rows lie inside the box.

There are at most γ active diagonals and at most γ active anti-diagonals. Each such line contains at most one square in column a and at most one in column b , and hence meets zero, one, or two squares of V . It is useful to think of one square per active diagonal or anti-diagonal as the baseline, corresponding to coverage without waste. Such a line that misses V wastes one unit, while one that meets V twice provides one extra unit that can compensate for such waste.

At the baseline, the rows can cover at most $2(\gamma - n + 1 + d - c)$ squares of V , while the active diagonals and anti-diagonals together contribute at most 2γ baseline units. Since V contains $2(d - c + 1)$ squares, this leaves only $2(\gamma - n + 1 + d - c) + 2\gamma - 2(d - c + 1) = 2(2\gamma - n) = 2\delta$ units available for waste. Each diagonal or anti-diagonal that meets V twice provides one additional unit. Thus, the total waste is limited by a small *budget* 2δ plus the number of such lines. The following lemma quantifies exactly how much waste can be afforded.

Lemma 1. *Let D be a dominating set of at most γ queens on the $n \times n$ board that leaves at least two rows and two columns empty. For the vertical sides V , let*

$$\text{over}(V) = \sum_{z \in V} (\text{number of active lines through } z - 1),$$

and let $\text{miss}(V)$ and $\text{twice}(V)$ denote, respectively, the number of active diagonals and anti-diagonals that meet V zero and two times. Define

$$e_{\text{row}} = \gamma - (\text{number of active rows}),$$

and define e_{col} , e_{diag} , and e_{anti} analogously. These excesses are non-negative, since at most γ queens can activate at most γ lines of each type. Define $\text{over}(H)$, $\text{miss}(H)$, and $\text{twice}(H)$ analogously for the horizontal sides H . Then

$$\text{over}(V) + \text{miss}(V) + 2e_{\text{row}} + e_{\text{diag}} + e_{\text{anti}} = \text{twice}(V) + 2\delta, \quad (1)$$

$$\text{over}(H) + \text{miss}(H) + 2e_{\text{col}} + e_{\text{diag}} + e_{\text{anti}} = \text{twice}(H) + 2\delta. \quad (2)$$

Proof. We count the active lines through the squares of V in two ways.

By squares. For each square of V , count the number of active lines through it. Since D is dominating and the two columns containing V are empty, every square has at least one such line. Counting one active line for each of the $2(d - c + 1)$ squares gives $2(d - c + 1)$, while all additional active lines are counted by $\text{over}(V)$. Thus, the total is $2(d - c + 1) + \text{over}(V)$.

By lines. The columns containing V are empty and therefore contribute nothing. There are $\gamma - e_{\text{row}}$ active rows in total. Of these, exactly $n - 1 - (d - c)$ lie outside the box, leaving $\gamma - e_{\text{row}} - (n - 1 - (d - c))$ active rows strictly inside the box. Each of these meets V twice, so the rows contribute $2(\gamma - e_{\text{row}} - n + 1 + d - c)$.

There are $\gamma - e_{\text{diag}}$ active diagonals and $\gamma - e_{\text{anti}}$ active anti-diagonals. Taking one square of V per active line as the baseline, a line that misses V contributes one fewer, while a line that meets V twice contributes one additional square. Thus, the diagonals and anti-diagonals contribute $(\gamma - e_{\text{diag}}) + (\gamma - e_{\text{anti}}) - \text{miss}(V) + \text{twice}(V)$.

Equating the two counts gives $2(d - c + 1) + \text{over}(V) = 2(\gamma - e_{\text{row}} - n + 1 + d - c) + (\gamma - e_{\text{diag}}) + (\gamma - e_{\text{anti}}) - \text{miss}(V) + \text{twice}(V)$. Rearranging and using $\delta = 2\gamma - n$ gives $\text{over}(V) + \text{miss}(V) + 2e_{\text{row}} + e_{\text{diag}} + e_{\text{anti}} = \text{twice}(V) + 2\delta$, which is (1). Equation (2) follows by the same argument with rows and columns exchanged. $\square$

For a fixed box, let T_V be the set of diagonals and anti-diagonals that contain two squares of V , and define T_H analogously for H . Unlike $\text{twice}(V)$ and $\text{twice}(H)$, these sets depend only on the box. Let L_ℓ indicate whether line ℓ is active. Then $\text{twice}(V)$ and $\text{twice}(H)$ count the active lines in T_V and T_H , respectively. Thus, $\text{twice}(V) = \sum_{\ell \in T_V} L_\ell$ and $\text{twice}(H) = \sum_{\ell \in T_H} L_\ell$. Consequently,

$$|T_V| - \text{twice}(V) = \sum_{\ell \in T_V} (1 - L_\ell) \quad \text{and} \quad |T_H| - \text{twice}(H) = \sum_{\ell \in T_H} (1 - L_\ell),$$

which count the lines that could meet the corresponding sides twice but are inactive. Moving $\text{twice}(V)$ and $\text{twice}(H)$ to the left in Equations (1) and (2), and adding $|T_V|$ and $|T_H|$, respectively, gives

$$\text{over}(V) + \text{miss}(V) + 2e_{\text{row}} + e_{\text{diag}} + e_{\text{anti}} + \sum_{\ell \in T_V} (1 - L_\ell) = |T_V| + 2\delta, \quad (3)$$

$$\text{over}(H) + \text{miss}(H) + 2e_{\text{col}} + e_{\text{diag}} + e_{\text{anti}} + \sum_{\ell \in T_H} (1 - L_\ell) = |T_H| + 2\delta. \quad (4)$$

In this form, the left sides consist entirely of waste, while the budgets on the right depend only on the box and δ . The main purpose of this rewrite is that the resulting constraints are easier to encode in SAT than their original forms.

By the geometry of the box, $T_V = \emptyset$ if the box is wider than tall, $T_H = \emptyset$ if it is taller than wide, and $|T_V| = |T_H| = 2$ if it is square.

Corollary 1. For $n \geq 3$, $\gamma(Q_n) \geq (n - 1)/2$.

Proof. Take a minimum dominating set and assume its box is at least as wide as tall (otherwise transpose the board). Then $\text{twice}(V) \leq |T_V| \leq 2$, and the left side of (1) is non-negative, so $2 + 2\delta \geq 0$, that is, $2\gamma - n \geq -1$. $\square$

When $\gamma = n/2$, we have $\delta = 0$, so the budgets reduce to $|T_V|$ and $|T_H|$. If the box is wider than tall, the budget for V is therefore zero, if it is taller than wide, the budget for H is zero, and when the box is square, both budgets are two. Thus, only a very small amount of waste is possible. In this sense, a dominating set of $n/2$ queens is virtually a set of non-attacking queens whose lines cover the sides of the box exactly once. Our main conjecture is that this rigidity is difficult for a SAT solver to discover on its own, and that efficiently encoding this counting argument may provide useful hints to the solver. Thus, the rest of the work seeks an efficient encoding of this argument and investigates its impact on solver performance empirically.

3 SAT-Based Methodology

For an $n \times n$ board and a number of queens γ , we generate one formula for every possible box, which decides whether γ queens can dominate the board with that fixed box. Every dominating set has exactly one box, so the formulas for different boxes describe disjoint sets of solutions and together cover all possible solutions. Some boxes can be eliminated before generating the formulas. In particular, boxes with more than γ columns outside or more than γ rows outside are impossible, since each such column or row requires its own queen. Moreover, reflecting or transposing the board maps symmetric boxes to each other, so we keep only one box from each symmetry class. Each remaining formula consists of a core encoding of domination in terms of squares and lines, additional clauses immediately implied by the fixed box, and the counting hints from Section 2.

3.1 Domination encoding

For every square i , let Q_i indicate whether a queen is placed on i , and for every line ℓ , let L_ℓ indicate whether ℓ is active. Let $L(i)$ denote the set consisting of the row, column, diagonal, and anti-diagonal passing through square i . Our encoding is based on the line-variable encoding of Rostami and Bright [8]. For every line ℓ , we add

$$\neg L_\ell \vee \bigvee_{i \in \ell} Q_i,$$

so that an active line contains a queen. For our counting constraints, we also need the converse relation. Thus, for every square i and every $\ell \in L(i)$, we add

$$\neg Q_i \vee L_\ell.$$

Consequently, L_ℓ is true exactly when ℓ contains a queen. Since a square is dominated whenever at least one of its four lines contains a queen, for every square i we add

$$\bigvee_{\ell \in L(i)} L_\ell.$$

To bound the size of the dominating set, we impose

$$\sum_i Q_i \leq \gamma.$$

Moreover, since each queen belongs to exactly one row, one column, one diagonal, and one anti-diagonal, at most γ lines of each type can be active. Thus, writing $\mathcal{R}$, $\mathcal{C}$, $\mathcal{D}$, and $\mathcal{A}$ for the sets of rows, columns, diagonals, and anti-diagonals, respectively, we impose

$$\sum_{\ell \in \mathcal{R}} L_\ell \leq \gamma, \quad \sum_{\ell \in \mathcal{C}} L_\ell \leq \gamma, \quad \sum_{\ell \in \mathcal{D}} L_\ell \leq \gamma, \quad \sum_{\ell \in \mathcal{A}} L_\ell \leq \gamma.$$

3.2 Box encoding

For a fixed box with empty columns $a < b$ and empty rows $c < d$, we first force its four sides to be empty. Thus, for every square i in column a , column b , row c , or row d , we add

$$\neg Q_i.$$

Every column to the left of a or to the right of b , and every row below c or above d , must be active. For each such line ℓ , we add

$$\bigvee_{i \in \ell} Q_i \quad \text{and} \quad L_\ell.$$

3.3 Counting hints

We now encode the counting constraints (3) and (4). Since the box is fixed, we can determine in advance which diagonals and anti-diagonals miss each pair of sides and which meet them twice. For each side, every term on the left-hand side is represented by literals that become true when the corresponding waste occurs. We then constrain the number of these literals by the budget on the right-hand side.

Over-coverage. For every square z of V , let $L_V(z)$ be the set of lines through z other than the empty column containing it. Thus, $|L_V(z)| = 3$, except at the corners of the box, where the empty row is also excluded and $|L_V(z)| = 2$. We introduce a variable $o_{z,2}$ that is forced to be true whenever at least two lines in $L_V(z)$ are active. For every pair $\ell, \ell' \in L_V(z)$, we therefore add

$$L_\ell \wedge L_{\ell'} \rightarrow o_{z,2}.$$

If $|L_V(z)| = 3$, say $L_V(z) = \{\ell_1, \ell_2, \ell_3\}$, we additionally introduce $o_{z,3}$ and add

$$L_{\ell_1} \wedge L_{\ell_2} \wedge L_{\ell_3} \rightarrow o_{z,3}.$$

Thus, a square covered by two active lines contributes one over-coverage literal, while a square covered by three active lines contributes two. These literals together count $\text{over}(V)$. The corresponding literals for H are defined analogously.

Missing lines. For a fixed box, we know which diagonals and anti-diagonals do not meet V . For each such line ℓ , the variable L_ℓ itself serves as a waste literal, since it contributes to $\text{miss}(V)$ exactly when the line is active. The same construction is used for H .

Excess. The row excess is at least j exactly when at least $n - \gamma + j$ rows are inactive. We therefore use a literal for each such threshold, so that the number of true threshold literals represents the row excess. The column, diagonal, and anti-diagonal excesses are encoded analogously.

Final constraints. Recall that T_V is the set of diagonals and anti-diagonals that can meet V twice. For each $\ell \in T_V$, the literal $\neg L_\ell$ contributes one unit of waste in (3). We therefore collect the over-coverage literals for V , the active-line literals for diagonals and anti-diagonals that miss V , the row-excess literals twice, the diagonal- and anti-diagonal-excess literals once, and $\neg L_\ell$ for every $\ell \in T_V$. We impose

$$\sum_{w \in W_V} w \leq |T_V| + 2\delta,$$

where W_V denotes this multiset of waste literals. Listing the row-excess literals twice gives them weight two without requiring a weighted pseudo-Boolean constraint. The constraint for H is constructed analogously, with column excess in place of row excess:

$$\sum_{w \in W_H} w \leq |T_H| + 2\delta.$$

4 Implementation

We generate the formulas in Python using PySAT’s totalizer-based cardinality encodings. For each box, CaDiCaL solves the formula, and `drat-trim` checks its DRAT proof of unsatisfiability. We independently verify that the generated boxes cover all boxes up to symmetry. We validated the implementation against known domination numbers, confirming that $\gamma(Q_n) - 1$ queens cannot dominate the corresponding boards. Also, all satisfying models are checked by an independent verifier. To check that the counting hints preserve all solutions, we enumerated minimum dominating sets for $n = 4, 5, 6, 7, 8, 9, 10, 12, 13, 17$, and 18. The respective counts up to symmetry are 3, 37, 1, 13, 638, 21, 1, 1, 41, 22, and 2, agreeing with Rostami and Bright [8]. The implementation is available at <https://github.com/TahaRostami/Gamma/tree/hint-and-conquer>.

5 Results

For even n , the lower bound $\gamma(Q_n) \geq (n - 1)/2$, found independently by Raghavan and Venkatesan [7] and by Spencer (see [2]), gives $\gamma(Q_n) \geq n/2$. On the other hand, the dominating sets constructed by Östergård and Weakley [6] show that $\gamma(Q_n) \leq n/2 + 1$ for every even n with $26 \leq n \leq 120$. Hence, for these n the domination number is either $n/2$ or $n/2 + 1$, and apart from $\gamma(Q_{30}) = 15$ the exact value was not known [10]. To settle such a case, it suffices to decide whether $n/2$ queens can dominate the board. We answer this question for $n = 26, 28, 32, 34, 36$, and 38.

The experiments of this section were carried out on the Iris HPC cluster at the University of Luxembourg, using Slurm job arrays on compute nodes equipped with either two Intel Xeon E5-2680 v4 (2.4 GHz) or two Intel Xeon Gold 6132 (2.6 GHz) processors.

n	γ	boxes	conflicts	solve	check	proof	$\gamma(Q_n)$
26	13	1596	3,188,497	4.4	5.2	799.3 MB	14
28	14	2080	4,722,892	6.8	7.6	1.18 GB	15
32	16	3321	9,879,747	14.8	15.7	3.06 GB	17
34	17	4095	15,095,776	23.6	23.5	5.28 GB	18
36	18	5050	26,679,740	34.7	40.1	8.59 GB	19
38	19	6105	40,968,906	63.0	88.3	13.13 GB	20

Table 1: The new instances. For each of them, every box is unsatisfiable and every proof is checked, so γ queens do not dominate the $n \times n$ board. The solving time and the proof-checking time are given in minutes. Conflicts, times, and proof sizes are summed over all boxes.

Table 1 lists the new instances. In every row all boxes are unsatisfiable, so $n/2$ queens do not dominate the $n \times n$ board and $\gamma(Q_n) \geq n/2 + 1$. Together with the upper bound above, we obtain the new values

$$\gamma(Q_{26}) = 14, \quad \gamma(Q_{28}) = 15, \quad \gamma(Q_{32}) = 17, \quad \gamma(Q_{34}) = 18, \quad \gamma(Q_{36}) = 19, \quad \gamma(Q_{38}) = 20.$$

Together with the previously known values, $\gamma(Q_n)$ is now known for every even $n \leq 38$. The largest instance, $n = 38$, needs 63.0 minutes of solving and 88.3 minutes of proof checking.

6 Comparison

To measure how much of the performance is due to the counting hints, we compare three configurations on board sizes whose domination number is known. The first is the full encoding of Section 3. The second, *no hints*, is the same encoding without the counting hints of Section 3.3. The third is the encoding of Rostami and Bright [8], the previous state-of-the-art SAT encoding for queen domination.

For each board size n , we ask whether $\gamma(Q_n) - 1$ queens can dominate the board, so every instance is unsatisfiable. The experiments of this section were run on a machine with an Intel Core i7-13700H processor (14 cores, 20 logical processors, base frequency 2.40 GHz).

Table 2 shows the results. Removing the hints makes every instance slower. Compared to the encoding of Rostami and Bright [8], the full encoding is 4.0, 3.8, and 3.3 times faster for $n = 14, 15$, and 16 , respectively. For $n = 19$, their encoding does not finish within one hour, while the full encoding needs 12.28 seconds.

n	γ	δ	full encoding	no hints	Rostami and Bright [8]
14	7	0	3.34	22.7	13.48
15	8	1	49.63	376.9	190.62
16	8	0	8.36	138.2	27.79
19	9	-1	12.28	243.31	timeout

Table 2: Time in seconds to show that $\gamma = \gamma(Q_n) - 1$ queens do not dominate the $n \times n$ board.

The split into boxes does not explain the improvement by itself. For $n = 14, 15$, and 16 , the configuration without hints is slower than the encoding of Rostami and Bright [8], by a factor between 1.7 and 5.0. Only for $n = 19$ does it finish where their encoding does not. Thus, on the three smaller boards the whole gain comes from the counting hints, and the split is useful mainly because it fixes the box that the hints refer to.

The case $n = 19$ is the one in which the hints are strongest. With 9 queens we have $\delta = -1$, so the bounds in the two final constraints are $|T_V| - 2$ and $|T_H| - 2$. If the box is not square, one of T_V and T_H is empty, so the corresponding bound is -2 and the formula is unsatisfiable by this constraint alone, without any search. This refutes 395 of the 465 boxes. For the remaining square boxes both bounds are zero, which is not a contradiction by itself, and these are the only formulas on which the solver has to search. Accordingly, the solver needs only 2,381 conflicts in total for $n = 19$, compared to 308,062 for $n = 16$. Conversely, $n = 15$ is the only instance with $\delta = 1$, hence with the largest budgets, and it is the slowest instance for the full encoding, with 2,179,963 conflicts.

Note that the comparison is restricted to this task. For enumerating minimum dominating sets, for which the encoding of Rostami and Bright was designed, we do not claim an improvement.

7 Conclusion

In this paper, we determine six new queen domination numbers, namely $\gamma(Q_n) = n/2 + 1$ for $n = 26, 28, 32, 34, 36$, and 38 . For each of these boards, the domination number was known to be either $n/2$ or $n/2 + 1$, and we show that $n/2$ queens are insufficient. As a result, the domination number is now known for every even $n \leq 38$. We achieve this through a SAT-based approach that partitions the search space according to the box defined by the outermost empty rows and columns of a dominating set, and solves one formula for each box.

The key ingredient of our approach is a set of counting hints. They are derived from the classical counting argument behind the lower bound $\gamma(Q_n) \geq (n - 1)/2$, which we refine into

an identity that limits how much coverage of the sides of the box can be wasted. Once the box is fixed, this identity is encoded with two cardinality constraints over the line variables of the encoding of Rostami and Bright [8]. In our experiments, the counting hints substantially improve solving performance compared with the same encoding without hints. For proving that a given number of queens is insufficient, the resulting encoding also outperforms that of Rostami and Bright on the tested instances.

Furthermore, our approach generates DRAT proofs for unsatisfiable formulas, which are independently verified using `drat-trim`. Consequently, the correctness of our unsatisfiability results depends on the correctness of the SAT encoding and the proof checker, rather than on the correctness of the SAT solver itself.

More broadly, this work demonstrates that counting arguments from the mathematical literature can be turned into effective hints for SAT solvers. Such arguments are difficult for a solver to derive on its own, but stating them explicitly can make a substantial difference in solving performance. The ideas developed in this work, namely splitting the search space so that a counting argument applies to each part, and encoding that argument as redundant constraints, may also prove useful for the remaining open cases of the queen domination problem, for other graph domination problems and, more generally, for proof-producing SAT applications.

References

- [1] W. H. Bird. *Computational methods for domination problems*. PhD thesis, University of Victoria, 2017.
- [2] E. J. Cockayne. Chessboard domination problems. *Discrete Mathematics*, 86:13–20, 1990.
- [3] P. B. Gibbons and J. A. Webb. Some new results for the queens domination problem. *Australasian Journal of Combinatorics*, 15:145–160, 1997.
- [4] M. D. Kearse and P. B. Gibbons. Computational methods and new results for chessboard problems. *Australasian Journal of Combinatorics*, 23:253–284, 2001.
- [5] Juho Lauri. Improved bounds on the domatic numbers of queens graphs. *Discrete Mathematics Letters*, 16:94–99, 2025.
- [6] P. R. J. Östergård and W. D. Weakley. Values of domination numbers of the queen’s graph. *Electronic Journal of Combinatorics*, 8(1):R29, 2001.
- [7] V. Raghavan and S. M. Venkatesan. On bounds for a board covering problem. *Information Processing Letters*, 25(5):281–284, 1987.
- [8] Taha Rostami and Curtis Bright. Queen domination by SAT solving. *The Art of Discrete and Applied Mathematics*, 2026. Accepted manuscript, published online 10 September 2026. Preprint: arXiv:2508.11945.
- [9] Bernardo Subercaseaux, Benjamin Przybicki, and Marijn J. H. Heule. Coloring queens with thousands of encodings, 2026. arXiv:2609.27674. To appear in LPAR 2026.
- [10] W. D. Weakley. Queen domination of even square boards. *Electronic Journal of Combinatorics*, 29(2):P2.50, 2022.